

Cohesion Policy and Institutional Quality Impact: Does Urban–Rural Classification Matter?

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Abstract

This study examines the impact of Cohesion Policy, and institutional quality, on regional economic performance across NUTS2 regions in 27 EU member states from 2007 to 2022. Average per capita Cohesion Policy payments are much higher in less developed regions, showing the policy focus on supporting lagging areas. Furthermore, the analysis groups regions as urban, rural and intermediate at the NUTS2 level to reflect key territorial differences and better assess how Cohesion Policy impacts vary across distinct development challenges. However, very low institutional quality in these regions may limit the effectiveness of this financial support. Using a two-step system generalised method of moments approach, the analysis finds that Cohesion Policy transfers have a positive impact on regional economic performance, with notable variation across regional types. The interaction between regional typologies and funding indicates that the policy is particularly effective in less developed, rural and intermediate regions, where structural disadvantages are more pronounced. Institutional quality plays a moderating role in this relationship and emerges as a key factor in enhancing Cohesion Policy effectiveness. To increase the impact of Cohesion Policy in lagging and rural regions, financial support should be complemented with efforts to improve institutional quality.

Keywords: EU cohesion policy; institutional quality; less developed regions; urban/rural regions

Introduction

The primary aim of the EU Cohesion Policy (CP) is to address regional disparities and foster economic development across its member states. Over the past four decades, the policy has evolved to become a cornerstone of the EU's efforts to achieve territorial and economic integration, currently allocating approximately 35% of the EU budget to initiatives aimed at reducing development differences between regions. From the start, the policy has focused on lagging regions, those with GDP per capita below 75% of the EU average, aiming to reduce development gaps and boost competitiveness (Cerqua and Pellegrini, 2017; European Commission, 2021). The CP supports infrastructure, socio-economic and capacity-building projects to promote equitable growth. It also helps integrate disadvantaged regions into the broader EU framework, contributing to the Union's long-term goals of inclusive and balanced development (Bachtler et al., 2016).

Evidence suggests that some regions use CP funds less effectively when they receive more money (Becker et al., 2012, 2008). This is often linked to weak institutions and low administrative capacity. Poor-quality institutions can slow down development and reduce the impact of regional policies. Many studies and the European Commission highlight that strong institutions are a key to economic growth and for making CP more effective (Charron et al., 2010, 2014; Ederveen et al., 2006; Rodríguez-Pose and Ketterer, 2020).

In less developed regions (LDR), weak institutions are a major obstacle. They limit the proper use of funds and reduce the potential benefits of CP. Good local governance helps co-ordinate policies, deliver services and align goals with local needs (Rodríguez-Pose and Garcilazo, 2015). Differences in institutional quality across regions help explain why CP works better in some regions than others (Aiello et al., 2019; Brasili et al., 2023; Quintieri and Stamato, 2021). Research shows that regions with strong institutions are better at managing funds, fighting corruption and promoting sustainable growth (Charron et al., 2014; Crescenzi and Giua, 2016).

More recent research has expanded on these findings. For regions receiving high levels of CP funds, improvements in government quality are more important for sustained economic growth than merely increasing investment (Rodríguez-Pose and Garcilazo, 2015). Similarly, Barbero et al. (2022) demonstrated that improving regional government quality could enhance the short-term and long-term impact of Cohesion investments by up to 7% and 3%, respectively. Similarly, Tiganasu and Lupu (2023) found that institutional quality and digitalisation played a crucial role in fund absorption in Central and Eastern European regions, which, in turn, contributed to regional development and reduced disparities. These findings suggest that addressing institutional weaknesses should be a priority to maximise the effectiveness of EU regional policies.

Another factor that affects both economic development and institutional quality is specific territorial configuration as Predominantly Urban, Predominantly Rural, and Intermediate regions, which may lead to differences in how CP performs. This classification is important for understanding the diverse socio-economic conditions, development needs and institutional capacities of regions. The 2022 Annual Report on the State of the Regions and Cities in the European Union (European Committee of the Regions, 2022) highlights the most pressing issues facing European regions and cities, confirming a significant territorial divide between rural and urban areas. Rural populations are rapidly shrinking, whilst the elderly population grows. The report further emphasises that long-term economic decline is especially severe in rural areas. Urban areas often enjoy stronger economic activity, better infrastructure and more effective governance, whilst rural regions typically face challenges such as low population density (PD), weaker institutions and limited resources, which can reduce the effectiveness of CP (Bachtler et al., 2016; Dijkstra and Poelman, 2014). Intermediate regions combine both urban and rural features, presenting unique opportunities and constraints. For example, rural regions located near cities may benefit more from CP due to their proximity to urban markets and infrastructure (Gagliardi and Percoco, 2017). However, many rural regions still lag, despite receiving significant CP funding, highlighting the need to better tailor interventions to local conditions (Egri and Lengyel, 2024). This typology is therefore useful for examining how CP interacts with regional characteristics to influence overall economic performance (Crescenzi and Giua, 2020). Understanding these differences can help policy-makers design more effective and equitable policies.

This study examines how CP affects regional economic performance by focusing on both the quality of local institutions and territorial characteristics. A novel and central part of the analysis is the classification of NUTS2 regions into Predominantly Urban, Predominantly Rural and Intermediate categories based on the share of the rural population, following OECD criteria. Compared to previous regional classification efforts, such as Gagliardi and Percoco (2017), who used proximity to urban centres at the NUTS3 level,

Brezzi et al. (2011), who added spatial accessibility criteria, and Egri and Lengyel (2024), who focused on spatial population patterns, our approach prioritises demographic structure. By focusing on the share of the rural population, our classification is more aligned with the territorial focus of CP. This ensures a better fit with the CP goals of promoting balanced regional development. Although this classification is more commonly applied at the NUTS3 level, adapting it to NUTS2, the main level for CP funding and implementation, provides a more policy-relevant framework for analysis. This approach allows us to explore how the impact of CP differs across types of regions that face very different development challenges, from low PD and infrastructure needs in rural areas to issues like congestion and housing pressure in urban ones. By capturing this diversity, the study contributes to understanding how policy can be better tailored to local needs, in line with the EU place-based approach to regional development (Barca, 2009).

In addition, the study integrates the European Quality of Government Index (EQI) at the NUTS2 level, offering a clearer picture of how differences in institutional quality, such as transparency, accountability and administrative capacity, shape the effectiveness of CP. Whilst existing studies emphasise the importance of institutional quality in mediating the success of Structural Funds (Charron et al., 2014; Rodríguez-Pose and Garcilazo, 2015), few have systematically explored how these dynamics vary across urban, rural and intermediate regions. Many studies rely on national-level indicators, whereas using region-specific governance data provides more precise insights. In addition to total CP funding, the study also examines the separate impacts of the European Regional Development Fund (ERDF) and the European Social Fund (ESF). To account for endogeneity and unobserved heterogeneity, key challenges in policy impact evaluations (Becker et al., 2018; Crescenzi and Giua, 2016), the study applies a two-step system generalised method of moments (GMM) approach. This helps generate robust causal evidence on how both CP funding and governance quality influence regional economic outcomes.

This study finds that CP funds significantly enhance regional economic performance. Disaggregated analysis of the ERDF and ESF confirms their positive effects, though the sign and magnitude of impact vary notably across regional typologies and development levels. For instance, economic performance in terms of per capita regional GDP is lower in less developed, predominantly rural and intermediate regions; however, the interaction of these regional typologies across both aggregated and disaggregated with CP funding shows that funds are particularly effective in stimulating economic performance in less developed, predominantly rural and intermediate regions, where structural disadvantages are more pronounced. Furthermore, the analysis highlights institutional quality as a key moderating factor; regions with higher governance capacity and transparency experience greater benefits from CP transfers. These findings highlight the importance of adapting CP interventions to the specific characteristics of each region. They also demonstrate that stronger institutional frameworks can help disadvantaged regions better use CP funds, leading to more balanced and sustainable economic convergence across the EU.

The structure of this article is as follows: Section I outlines the dataset and its key features as well as the model specifications and examines the suitable econometric method

used for its estimation. The empirical findings are presented and discussed in Sections II and III, respectively, and the article concludes with a summary in the Conclusion section.

I. Methodology and Data

Methodology

Three strands of literature exist that provide insights into the CP impact on regional economic performance, including the neo-classical growth model, endogenous growth models and the literature on economic geography. The neo-classical growth model, which is the most cited in this context, predicts convergence of income amongst regions because of diminishing marginal product of capital (Barro, 1991; Bosca et al., 1999; Solanes and María-Dolores, 2002). The endogenous growth model, on the other hand, focuses on the mechanisms (such as investing in human capital, innovating and enhancing total factor productivity) that allow public policies to influence long-run growth (Aschauer, 1989; Barro, 1990; Pereira, 1999). The effects of geographical location/proximity and the importance of spatial interdependence on the outcome variable are explained by the economic geography literature (Krugman, 1991).

We rely on the neo-classical framework and specify a dynamic panel data regression model. Following the literature (Bouayad-Agha et al., 2013; Mohl and Hagen, 2010; Rodríguez-Pose and Garcilazo, 2015, amongst others), we hypothesise that institutions at the regional level are a requirement for regions to foster economic development (Charron et al., 2010). The effectiveness of CP relies on regional institutional quality in shaping regional economic performance by influencing policy implementation, resource allocation and stakeholder co-ordination (Bachtler and Mendez, 2016). Strong local institutions play a role in the absorption of EU funds, in fostering innovation and in reducing disparities; on the other hand, weak institutions may lead to inefficiencies or misallocation (Augusztn et al., 2025). Empirical evidence suggests that regions with transparent governance and administrative capacity achieve better outcomes from CP investments (Rodríguez-Pose, 2013). To examine this, a dynamic panel data econometric model is used to analyse the link between CP, institutional quality and regional economic performance. The aim is to determine whether the benefits of CP depend on the quality of regional governance and vary across regions with different levels of development and urbanisation. We use data on 240 NUTS2 regions covering the period from 2007 to 2022.

We start with the following specification as our baseline empirical model:

$$\ln(y_{it}) = \alpha + \beta \ln(y_{i,t-1}) + \delta \ln(CP_{it}) + \pi X_i + \epsilon_{it} \quad (1)$$

where $\ln(y_{it})$ is the natural logarithm of per capita GDP of region i at time t and $\ln(y_{i,t-1})$ is the one-period lag of per capita GDP in region i . The main variable of interest in Equation (1) is CP (and its components ERDF and ESF), which represent the per capita CP investment in region i at time t . We incorporate also a set of control variables represented by

the vector X that are essential for capturing dynamics of regional economic performance. Finally, the disturbance term is given by ϵ_{it} .

However, one of the main goals of this study is to examine both the direct effect of institutional quality and its interaction with CP funding on regional economic performance. To achieve this, we estimate an extended version of Equation (1) by including a measure of institutional quality and its interaction with CP:

$$\ln(y_{it}) = \alpha + \beta \ln(y_{i,t-1}) + \delta \ln(CP_{it}) + \varnothing EQI_{it} + \gamma \ln(CP_{it}) * EQI_{it} + \pi X_i + \epsilon_{it} \quad (2)$$

The measure of institutional quality, the moderator variable, is the EQI,¹ collected by the Quality of Government Institute at the University of Gothenburg; the interaction term between EQI and CP is given by $\ln(CP_{it}) * EQI_{it}$ and allows to measure the heterogeneity of CP impact by levels of EQI.

The coefficients such as δ , $\varnothing$ and γ are our main interest because they establish the relationship between per capita CP, EQI and the interaction of both terms, respectively, with regional economic performance.

Whilst much of the literature focuses on GDP growth rates (e.g., Barro, 1991), our approach is motivated by the need to understand how these factors influence the level of economic activity across regions (Acemoglu and Robinson, 2013; Glaeser et al., 1992; Henderson et al., 2018). The log transformation allows for an intuitive interpretation of coefficients in terms of percentage changes (Wooldridge, 2010) and aligns with theories of economic convergence, which emphasise reducing disparities in income levels (Barro and Sala-i-Martin 1992; Rodríguez-Pose and Fratesi, 2010). Furthermore, this approach is well suited to capturing the long-term effects of CP and institutional quality, which are more likely to influence the level of economic development than short-term growth dynamics (Bouayad-Agha et al., 2013; Fratesi and Rodríguez-Pose, 2016).

Different specifications are also evaluated, which in turn include some dummy variables. The dummy variable LDR represents the LDR. This dummy is computed by following the criteria used by the EU to target the LDR.² Moreover, dummy variables also account for the classification of NUTS2 regions into Predominantly Urban (hereafter PU), Predominantly Rural (hereafter PR) and Intermediate (hereafter IN) regions. These regional classifications are in line with standard practices for examining disparities in regional economic performance. Please refer to Appendix B for a detailed explanation of the methodology used for the classification of the NUTS2 regions into PU, PR and IN.

These dummy variables are crucial for capturing structural differences across regions, allowing the model to account for varying levels of economic development and urban–

¹The EQI is a measure of governance quality at the regional level within the EU, based on citizen surveys assessing public sector corruption, impartiality, and service quality. First introduced in 2010 and updated in 2013, 2017, 2021 and 2024, the EQI enables comparisons both within and across countries over time. The index serves as a valuable tool for researchers and policy-makers to analyse governance disparities and their implications for regional development. For detail on the EQI, please visit <https://www.gu.se/en/quality-government/qog-data/data-downloads/european-quality-of-government-index>.

²According to the criteria, a particular region is classified as less developed if GDP of that region is less than 75% of the EU27 average. Please see the data section for more information about the computation of this dummy.

rural characteristics. Such regional classifications have been employed by Gagliardi and Percoco (2017), who examined the differential impact of CP across regions, and Rodríguez-Pose and Garcilazo (2015), who emphasised the importance of geographical and economic heterogeneity in assessing the returns on CP investments.

To the specification of Equation (2) (Table 4, Model 2), we add a dummy variable for LDR and its interaction with CP payments to capture differences in regional economic performance between LDR and more developed regions (Table 4, Model 3). A fourth specification includes the dummy for PR regions and its interaction with CP payments to capture the differential impact of CP across PR and other regions (Table 4, Model 4). Finally, the dummy for IN and PR, together with their interaction with CP payments, is included in the last model (Table 4, Model 5).

System GMM was used to estimate all the models. The inclusion of the lagged dependent variable, $\ln(y_{i,t-1})$, as a regressor introduces endogeneity because it is correlated with the error term (Arellano and Bover, 1995). System GMM addresses these issues by leveraging internal instruments derived from lagged values of the endogenous variables, providing consistent and efficient parameter estimates (Blundell and Bond, 1998).

System GMM also allows for the inclusion of additional endogenous or predetermined variables, such as CP funds $\ln(CP_{it})$ and the interaction terms $\ln CP_{it} * dummy_{LDR/IN/PR}$. These variables are likely correlated with unobserved regional characteristics and contemporaneous shocks, which could lead to biased estimates if not properly addressed. System GMM effectively accommodates these complexities by using lagged values of the endogenous variables as instruments, ensuring robust inference.

Data and Variable Definitions

Our dataset covers the period 2007–2022, except for selected variables with different available years. GDP per capita in purchasing power standard (hereafter PPS) at the NUTS2 level represents the regional economic performance, retrieved from Eurostat regional statistics.

The CP transfer aims to reduce disparities, create stable economic conditions and improve the efficiency of the overall system by providing financial support to regions in the EU member countries (Molica et al., 2024). These transfers support EU member states and regions, particularly those with a GDP per capita below the EU27 average, aiming to foster socio-economic cohesion. The investment under the ERDF aims to make regions more competitive and smarter. It also focuses on making them greener, better connected through improved mobility and more socially inclusive by supporting employment, education and skills development (European Commission, n.d.). Similarly, the primary aim of the ESF is to improve the social dimension of development by investing in human capital, supporting job creation and inclusion and enhancing public services.

The CP variable measures the total EU annual payments per capita in Euros, which are distributed primarily through the ERDF, ESF and Cohesion Fund. Data are available from the European Commission (Eurostat) and the Cohesion Open Data Platform.

The EQI³ data for the years 2010, 2013, 2017 and 2021 for 206 NUTS2 (or NUTS1) regions are obtained from the QoG Institute University of Gothenburg.⁴ We perform interpolation only to fill missing values in the EQI; no extrapolation is applied. Since EQI data are available starting from 2010, we interpolate the missing values for the years from 2010 to 2021. However, we do not interpolate for the years 2007, 2008, 2009 and 2022, because interpolation only works within the range of available data points, not before or after them. This results in 960 missing values. Additionally, EQI data are only available for 206 out of the 240 NUTS2 regions included in our analysis. Therefore, interpolation is conducted only for those regions with at least some EQI data. This leads to an additional 544 missing values due to complete data unavailability in 34 regions over 16 years.

Regarding the regional characteristics (indicated by X in Equations (1) and (2)) that are essential for capturing the dynamics of regional economic performance, we consider:

- The employment rate (ER). The share of working age (15–64) population employed directly affects labour productivity and, as a result, regional output. This approach aligns with the findings of Becker et al. (2010), who demonstrate that labour market conditions are critical to assessing the impact of EU funds on growth.
- The PD. It is another important component of regional economic disparities, which, according to Crescenzi and Giua (2020), influences economies of scale, infrastructure development, and regional accessibility.
- The share of population with secondary school attainment (SSA). It is a proxy of human capital, reflecting the importance of education in driving productivity growth. Rodríguez-Pose and Garcilazo (2015) highlighted the role of human capital in determining the returns on CP and investment in LDR. Empirical studies exist that include only the share of the population with secondary education when examining its impact on regional economic performance or regional economic disparities. These studies often focus on the role of basic education and workforce skills as foundational drivers of economic performance, especially in less-developed regions or countries where tertiary education levels may be relatively low (Barro and Lee, 1994, 2013; Becker et al., 2010; Mankiw et al., 1992).
- Per capita gross fixed capital formation (GFCF). To capture the role of capital accumulation, we include per capita GFCF, a proxy for regional investments that is consistent to the framework applied by Mohl and Hagen (2010) in their analysis of EU Structural

³To address missing values in the EQI, which is observed only in 2010, 2013, 2017 and 2021, we apply linear interpolation for the intervening years. Let $EQI_{i,t}$ denote the EQI value for region i in year t . For any t such that $t \in (t_1, t_2)$, EQI_{i,t_1} and EQI_{i,t_2} are observed, the interpolated value is computed as:

$$EQI_{i,t} = EQI_{i,t_1} + \left(\frac{t - t_1}{t_2 - t_1} \right) \cdot (EQI_{i,t_2} - EQI_{i,t_1}).$$

This method assumes a linear change in EQI between two known time points and assigns proportionally weighted values to the intervening years. We do not perform extrapolation; therefore, no values are imputed for years prior to 2010 (i.e., 2007–2009) or after 2021 (i.e., 2022). Additionally, since EQI data are available for only 206 out of 240 NUTS2 regions, interpolation is carried out only for those regions with at least two observed data points. Regions with no EQI data throughout the observed period are excluded from this process.

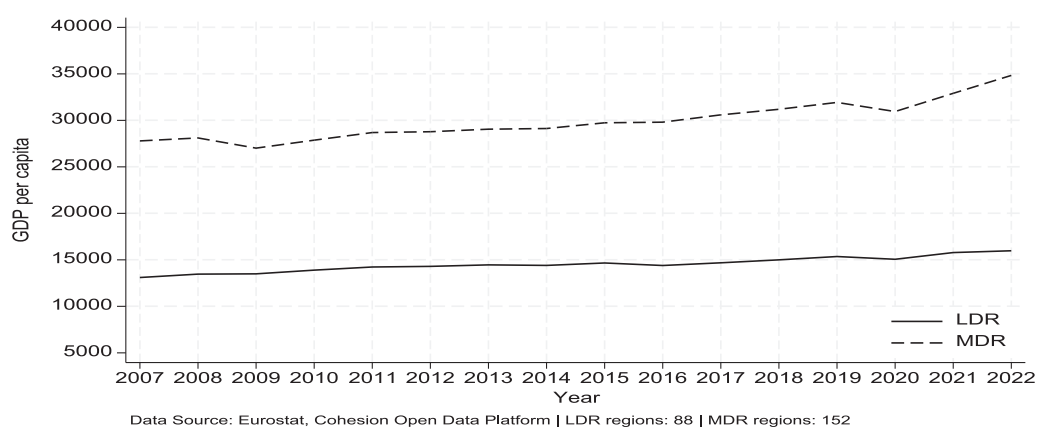
⁴Detail on the EQI can be found here: <https://www.gu.se/en/quality-government/qog-data/data-downloads/european-quality-of-government-index>.

Table 1: Distribution of NUTS2 Regions by Development Status and Geographic Characteristics.

Geographic characteristics	Development status		Total
	LDR	MDR	
Intermediate regions	38	74	112
Predominantly rural regions	24	40	64
Predominantly urban regions	26	38	64
Total	88	152	240

Notes: MDR stands for more developed regions and includes all the regions not classified as LDR.

Figure 1: Average GDP per Capita by Development Status (Euros).



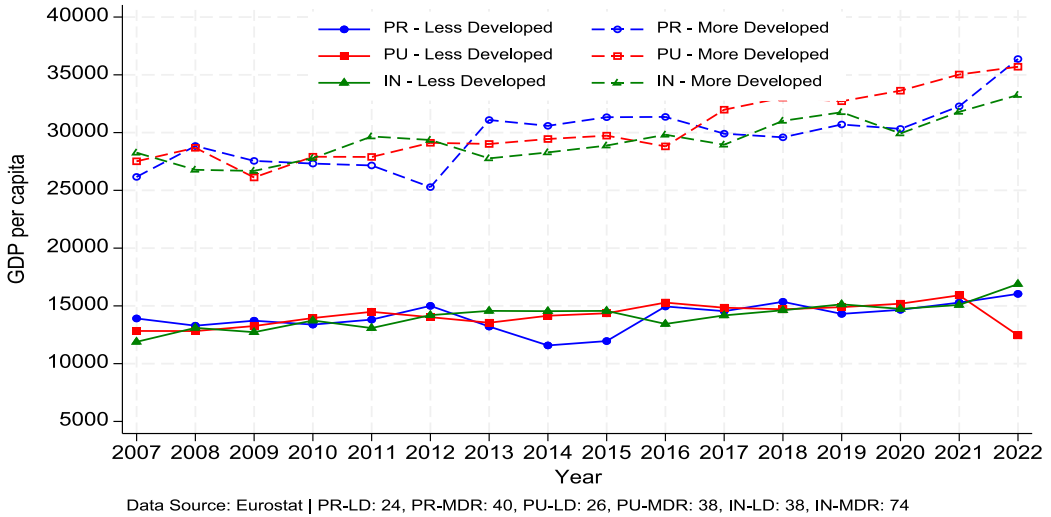
Funds. These data are sourced from the *General and Regional Statistics* database provided by Eurostat.⁵

Table 1 presents a summary of the regional classifications based on the geographical characteristics of the 240 regions considered. Out of the total 88 LDR, 26 are PU, 24 are PR and 38 are IN. Amongst the 152 more developed regions, 38 are PU, 40 are PR, whilst 74 are IN. In total, IN constitutes the biggest category, followed by PU and PR. This highlights the distribution of NUTS2 regions based on development status and geographic characteristics.

A detailed explanation and data source of the variables used in the study is provided in Appendix A.

⁵<https://ec.europa.eu/eurostat/data/database>.

Figure 2: Average GDP per Capita by Regional Classification and Development Status (Euros). [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/jcms.70050)]



II. Empirical Analysis

Descriptive Analysis

This section provides descriptive statistics for the key variables used in the analysis. The descriptive analysis is based on the data for the whole sample as well as for the different classifications of NUTS2 regions. First, by following EU criteria, we identify NUTS2 regions as LDR where their GDP is below 75% of the EU27 average (Eurostat, 2016).

Figure 1 provides a comprehensive overview of the economic performance of LDR and more developed regions (MDR) from 2007 to 2022, where MDR includes all the other regions. The figure reveals that whilst both groups experienced steady, modest growth during this period, MDR consistently maintained higher GDP levels, reflecting their stronger economic standing. Throughout the entire period, there is also a slight but clear increase in the gap between LDR and MDR, especially in the years following the COVID-19 pandemic.

Figure 2 illustrates GDP per capita trends from 2007 to 2022 across different regional classifications and development statuses. More developed urban regions (PU-MDR) consistently have the highest GDP per capita compared to PR-MDR and IN-MDR from 2016 until 2021, whilst LDR (PR-LD, PU-LD, IN-LD) exhibit significantly slower growth but not different trends across PR, PU and IN regional classification. The period before 2014 shows fluctuations, likely due to the 2008 Global Financial Crisis and the Eurozone Sovereign Debt Crisis (2010–2012), which caused economic instability and slow recovery in many regions. However, a steady upward trend emerges afterwards, with a sharp increase post-2020, possibly reflecting post-pandemic economic recovery efforts. Overall, the

Table 2: Descriptive Statistics for Key Variables by Selected Years.

Variable	Year	Whole sample		LDR	
		Mean	Std. dev.	Mean	Std. dev.
Per capita GDP (euros)	2007	23,631	9406	14,502	4079
	2015	26,285	10,475	15,617	3035
	2022	33,737	13,193	17,395	2387
Per capita CP (euros)	2007	12.83	18.27	22.54	20.87
	2015	66.40	98.22	130.98	115.39
	2022	165.80	425.69	103.02	142.96
Per capita ERDF (euros)	2007	8.50	12.37	16.25	15.80
	2015	50.68	84.54	102.63	100.77
	2022	117.01	311.73	69.01	90.56
Per capita ESF (euros)	2007	4.52	8.91	6.49	6.43
	2015	16.04	28.70	28.56	33.85
	2022	49.00	128.54	34.51	53.81
EQI	2015	−0.08	1.01	−0.86	0.56
Employment rate (%)	2007	69.15	6.95	63.92	6.71
	2015	69.14	8.62	62.00	8.26
	2022	74.83	7.82	69.60	6.76
Secondary school (%)	2007	48.33	16.47	54.04	19.59
	2015	48.42	13.89	51.87	16.26
	2022	46.77	12.45	54.85	9.23
Population density	2007	334.49	776.54	123.81	156.52
	2015	345.38	853.23	206.02	777.18
	2022	352.55	863.52	95.20	153.32
Per capita GFCF at constant prices in EUR 2015	2007	6130.63	3048.79	2645.12	1384.42
	2015	5502.54	3314.66	2070.10	672.09
	2022	13,224.69	32,751.18	1558.44	976.23

figure highlights persistent regional disparities, emphasising the need for policies to support economic performance in PR and IN regions.

Table 2 presents descriptive statistics for the key variables used in the study for the whole sample as well as for LDR. Considering the whole sample, the mean and standard deviation of per capita GDP and CP payments generally increase, reflecting an increase in the volume of these variables on one hand and uncertainty on the other. However, there are modest changes in the average value and uncertainty over time in other variables such as ERs, PD and SSA.

It is important to note that the average per capita CP payments are much higher across LDR as compared to the whole sample, reflecting the regional distribution of Structural Funds as the goal of CP to improve economic performance in lagging regions. Another important factor to note is that the average EQI in the whole sample is -0.083 , whilst the same in the sample of LDR is -0.859 .⁶ Whilst the MDR's EQI is close to the average, the LDR's EQI is considerably lower, potentially hindering their ability to meet CP goals (as discussed in the [Regression Analysis](#) and [The Impact of CP and the Role of](#)

⁶The EQI ranges between -2.5 and 2.5 , and 0 is the EU-27 average.

Table 3: Summary Statistics of Key Variables for PU, PR and IN.

Variables	Years	PU regions		PR regions		IN regions	
		Mean	SD	Mean	SD	Mean	SD
Per capita GDP (euros)	2007	23,158	9906	22,911	7636	10,942	15,802
	2015	26,372	10,413	26,489	10,515	9909	9894
	2022	34,731	13,674	34,844	15,101	7581	1826
Per capita CP (euros)	2007	14.08	19.50	13.07	16.20	11.98	18.77
	2015	70.46	107.26	50.71	62.76	65.30	95.08
	2022	154.30	462.37	187.32	442.38	155.36	399.83
Per capita ERDF (euros)	2007	9.99	13.62	8.46	10.11	7.67	12.82
	2015	54.58	96.48	37.58	53.39	49.40	78.21
	2022	109.43	341.86	126.33	283.22	113.62	320.26
Per capita ESF (euros)	2007	4.28	6.16	4.83	9.10	4.50	10.12
	2015	16.25	32.56	13.44	13.44	16.22	27.13
	2022	45.10	122.96	61.22	173.92	41.96	86.50
EQI	2015	0.22	1.07	0.13	0.89	−0.03	0.96
Employment rate (%)	2007	68.58	7.77	68.46	6.97	69.90	6.42
	2015	69.84	8.55	69.94	9.83	68.54	8.56
	2022	76.48	6.17	74.44	8.27	74.44	8.10
Secondary school (%)	2007	48.24	17.25	48.61	17.62	48.22	15.45
	2015	49.87	13.21	46.67	13.31	47.57	14.45
	2022	49.14	11.72	45.45	12.71	46.73	12.54
Population density	2007	202.89	219.28	414.88	878.04	366.41	913.67
	2015	426.74	1064.19	182.15	166.71	305.42	716.14
	2022	290.62	456.69	371.38	972.65	365.66	917.51
Per capita GFCF at constant prices in EUR 2015	2007	5818.33	3085.11	5975.89	2692.89	6394.73	3218.46
	2015	5508.45	3354.46	5833.31	3254.24	5456.80	3315.80
	2022	16,951.98	53,946.73	15,548.75	34,465.99	9958.819	14,301.88

Institutional Quality sections). It is also important to note that the high average per capita GFCF in 2022 is primarily driven by a few highly developed regions with exceptionally large investments, whilst most LDR report much lower values. This results in a skewed distribution where the mean is inflated by outliers.

Table 3 breaks down the whole sample by region type into PU, PR and IN to capture economic disparities across these regions. The average per capita GDP of PU and PR increases over time, whilst it decreases in the case of IN. There is an increase in the per capita CP payments and their components over time in all three groups of regions. EQI is, as usual, higher in PU, followed by PR and then IN, reflecting weaker governance, transparency and institutional ineffectiveness in PR and IN regions. Without improvement in institutional quality, especially in LDR and PR regions, the effort to achieve the goals of CP seems useless. This calls for the authority to complement the financial support of the regions through CP with improvements in institutional quality. Notably, the unusually high average per capita GFCF in 2022 is driven by a few highly developed regions with large investments, whilst most LDRs report much lower values. This skewed distribution suggests that the median may better capture the central tendency than the mean.

Table 4: Cohesion Policy Payment and Regional Economic Performance.

Variables	<i>Model 1</i>	<i>Model 2</i>	<i>Model 3</i>	<i>Model 4</i>	<i>Model 5</i>
$(t - 1)$ log of per capita GDP	0.753*** (0.0281)	0.431*** (0.0553)	0.128*** (0.017)	0.371*** (0.0605)	0.656*** (0.0593)
Log of per capita CP	0.0247*** (0.00127)	0.0398*** (0.00331)	0.0360* (0.0197)	0.0527*** (0.0137)	0.0163* (0.00993)
EQI		0.0534*** (0.0194)	0.123** (0.0516)	0.0607** (0.0291)	0.0474** (0.0210)
Interaction CP*EQI		0.0301*** (0.00571)	0.0225** (0.00903)	0.0216*** (0.00710)	0.00744 (0.00683)
LDR			— 0.773*** (0.221)		
Interaction CP*LDR			0.0990** (0.0473)		
PR regions				— 0.373*** (0.0346)	— 0.0708** (0.0328)
Interaction CP*PR				0.0141*** (0.0020)	0.0321* (0.0180)
IN regions					0.0687** (0.0318)
Interaction CP*IN					0.0351** (0.0168)
Log of Employment rate	0.172*** (0.0368)	0.488*** (0.0665)	0.186** (0.090)	0.447*** (0.0796)	0.420*** (0.0724)
Log of Population density	0.0190*** (0.00367)	0.0353*** (0.00902)	0.0397** (0.0163)	0.0374*** (0.00992)	0.0172*** (0.00628)
Log of Secondary education	— 0.000101 (0.00952)	0.0613** (0.0245)	0.3940*** (0.0448)	0.0429* (0.0248)	— 0.0405** (0.0189)
Log of per capita GFCF	0.106*** (0.0139)	0.289*** (0.0279)	0.197*** (0.0550)	0.278*** (0.0326)	0.185*** (0.0326)
Constant	2.246*** (0.287)	5.309*** (0.635)	8.973*** (1.210)	5.966*** (0.672)	2.746*** (0.601)
Observations	3241	2091	2091	2091	2091
Number of NUT2	238	179	179	179	179
Number of instruments	35	54	33	33	35
Arellano–Bond test for AR(2) (<i>p</i> -value)	0.715	0.290	0.364	0.360	0.601
Hansen test of overid. restrictions (<i>p</i> -value)	0.606	0.115	0.126	0.168	0.291

Notes: ***, ** and * denote significance at the 1%, 5%, and 10% levels, respectively. The standard errors in parentheses are corrected standard errors. Arellano–Bond AR(2) test: H_0 : No AR(2) in residuals. Rejection $\Rightarrow$ autocorrelation, model misspecification. Hansen test: H_0 : Instruments are valid. Rejection $\Rightarrow$ instruments may be invalid.

Regression Analysis

The Impact of CP and the Role of Institutional Quality

To analyse the effect of CP on per capita GDP of 240 NUTS2 regions, we estimate different specifications of Equations (1) and (2) by two-step system GMM following a panel

data approach. This approach is relevant for dynamic panel models, which address endogeneity through lagged dependent variables as instruments. Specification tests such as the Arellano–Bond AR(2)⁷ and Hansen test confirm the reliability of these instruments.⁸

We treat the lagged dependent variable (one-period lag of per capita GDP) and the key policy variables (total CP, ERDF, ESF and the interaction terms involving them) as endogenous, as they may be correlated with unobserved regional characteristics. Control variables, including ER, PD, secondary education, GFCF and regional classification dummies (PU, PR and IN), are treated as predetermined or exogenous. To mitigate the risk of instrument proliferation, we implement instrument reduction techniques; that is, we use the collapse option and limit the number of lags for endogenous variables to a maximum of two (Bontempi and Mammi, 2012; Roodman, 2009). The total number of instruments remains well below the number of regions (238 in Model 1 and 179 in remaining models), with Model 1 using 35 instruments, Model 2 using 54, Model 3 and 4 using 33 and Model 5 using 35 instruments. The instrument-to-group ratio is thus kept safely below the critical 1:1 threshold, reducing the risk of overfitting.

To begin the analysis, we estimate the baseline⁹ model of Equation (1) that does not include the EQI. This is done as a robustness check to make sure that our results are not overly optimistic due to the limited availability of EQI data. Model 1 in Table 4 shows the results of this baseline model. The sample includes 238 NUTS2 regions and 3241 observations. The coefficient for the one-period lag of GDP per capita is positive and significant, which suggests that regions with higher GDP in the past tend to perform better in the present as well. The effect of CP payments is also positive and significant, meaning that these funds help improve regional economic performance. CP payments often support infrastructure, education and other development projects, which can increase productivity and attract investment.

In Model 2, to examine how EQI moderates the impact of CP on regional economic performance, we include EQI and their interaction. In this case, the sample comprises 179 NUTS2 regions due to missing data on EQI and a total of 2091 observations. Results suggest a complementary relationship between institutional quality and CP. In other words, this means that the impact of CP transfer is stronger in regions with better institutional quality, where transparent, efficient and accountable institutions improve planning, management and resource allocation as compared to those regions where EQI is low. Studies by Rodríguez-Pose and Garcilazo (2015) and Crescenzi and Giua (2016) show that well-governed regions maximise the returns on public investments, whilst poorly governed regions struggle due to inefficiencies and corruption. Similarly, Ederveen et al. (2006) and Becker et al. (2012) confirm that institutional quality enhances the effectiveness of CP.

⁷The Durbin–Wu–Hausman test to detect endogeneity bias is typically used in the context of linear instrumental variable regressions and is not directly applicable within the dynamic panel GMM framework.

⁸The Arellano–Bond AR(2) test shows no second-order serial correlation, supporting moment restriction validity. Similarly, the Hansen test indicates overidentifying restrictions are satisfied, confirming instrument validity and consistency in the GMM estimators used.

⁹Due to missing data, the inclusion of EQI reduces the estimation sample from 3241 to 2091 observations. To verify that differences in coefficients are not driven by this change in sample coverage, we also re-estimated Model 1 on the restricted sample (results are available from authors on demand). The results closely align with those from the full sample, confirming that the increase in the CP coefficient when EQI is introduced reflects the role of institutional quality rather than an artefact of sample reduction.

The results discussed above do not account for regional heterogeneity, in particular, the structural differences between LDR and MDR. Understanding their distinct impact is important because CP transfers (as remarked in the [Regression Analysis](#) section) are more heavily allocated to lower income regions (LDR). Therefore, to address this heterogeneity, we first introduce the dummy for LDR and its interaction with CP payments in Model 3, which allows us to examine whether the CP effect varies depending on the region development status.¹⁰ In this context, Model 3 (Table 4) shows that economic performance is lower in LDR compared to developed regions. The negative and significant coefficient for LDR indicates that LDR have lower economic performance, aligning with their need for development support (Barca et al., 2012; González-Val, 2015). However, the positive impact of the interaction between CP payments and LDR suggests that the CP can help mitigate these disadvantages by addressing structural constraints and stimulating growth. Studies' findings confirm that well-targeted CP payments have a more significant impact on LDR (Barca et al., 2012; Ederveen et al., 2006; González-Val, 2015; Mohl and Hagen, 2010). These results highlight the need for tailored CP to support marginalised regions, promoting balanced regional growth and reducing spatial inequalities across the EU regions.

Since regional disparities exist not only between developed and LDR but also between urban and rural areas, it is crucial to introduce additional layers of heterogeneity. Therefore, starting from Equation (2), we introduce the dummies PR and IN in the next two specifications (Models 4 and 5). These dummies and their interactions with CP account for how territorial characteristics (Urban vs. Rural and Urban vs. Intermediate regions) influence CP effectiveness. The effectiveness of CP may vary based on factors like infrastructure, PD and economic activity levels, which differ across these regional types. By integrating these spatial dimensions into the analysis, we aim to uncover whether the drivers of economic development vary systematically across urban, rural and intermediate regions, thereby providing insights into the role of regional context in shaping economic performance.

Results indicate that, compared to other regions, the economic performance of PR is lower (Model 4), suggesting that Predominantly Rural regions face structural disadvantages such as lower productivity and weaker infrastructure. Instead, intermediate regions perform even better than PU regions (Model 5). The positive interaction terms between CP payments and both PR and IN regions (Models 4 and 5) demonstrate that CP payments play a critical role in addressing these challenges, fostering economic development and reducing regional disparities. Moreover, the interaction of CP and EQI remains significant and positive, at least in Model 4, confirming that institutional quality enhances the effectiveness of CP. These results align with studies by Crescenzi and Giua (2016) and Rodríguez-Pose and Garcilazo (2015).

About the control variables, the positive and significant effects of ER, PD and GFCF on per capita GDP highlight their key role in regional economic performance. A higher ER boosts productivity, directly increasing per capita GDP. PD reflects urbanisation, where proximity to markets and infrastructure enhances economic output. The influence of the share of population with secondary education on regional economic performance is either insignificant or has a negative sign in Models 1 and 5. Empirical evidence on the

¹⁰ Models 3, 4 and 5 are estimated based on the same sample as Model 2.

Table 5: European Regional Development Fund and Regional Economic Performance.

	<i>Model 1</i>	<i>Model 2</i>	<i>Model 3</i>	<i>Model 4</i>	<i>Model 5</i>
$(t - 1)$ log of per capita GDP	0.732*** (0.0294)	0.518*** (0.0446)	0.395*** (0.0538)	0.436*** (0.0568)	0.521*** (0.0445)
Log of per capita ERDF	0.00805*** (0.00150)	0.00472** (0.00203)	0.0469*** (0.00490)	0.0246*** (0.00339)	0.0779** (0.0322)
EQI		0.0689*** (0.0119)	0.0923*** (0.0251)	0.0764*** (0.0186)	0.0624*** (0.0130)
Interaction ERDF*EQI		0.00700*** (0.00204)	0.0878*** (0.00601)	0.0595*** (0.00475)	0.00680*** (0.00211)
LDR			−0.0818* (0.0471)		
Interaction ERDF*LDR			0.0320*** (0.0118)		
PR regions				— 0.0798*** (0.0208)	−0.0315** (0.0121)
Interaction ERDF*PR				0.0596*** (0.00954)	0.0709** (0.02924)
IN regions					0.0139 (0.0234)
Interaction ERDF*IN					0.0685** (0.02984)
Log of Employment rate	0.186*** (0.0386)	0.410*** (0.0551)	0.443*** (0.0753)	0.467*** (0.0667)	0.391*** (0.0552)
Log of Population density	0.0219*** (0.00418)	0.0341*** (0.00646)	0.0354*** (0.00832)	0.0360*** (0.00791)	0.0351*** (0.00650)
Log of Secondary education	−0.000979 (0.0113)	0.0425** (0.0191)	−0.0373 (0.0254)	— 0.0496** (0.0235)	−0.0400** (0.0198)
Log of per capita GFCF	0.122*** (0.0148)	0.251*** (0.0218)	0.288*** (0.0304)	0.296*** (0.0293)	0.245*** (0.0233)
Constant	2.468*** (0.305)	4.488*** (0.488)	5.793*** (0.597)	5.342*** (0.633)	4.498*** (0.486)
Observations	3241	2091	2091	2091	2091
Number of NUTS2 regions	238	179	179	179	179
Number of instruments	35	54	33	33	56
Arellano–Bond test for AR(2) (<i>p</i> -value)	0.799	0.883	0.134	0.411	0.700
Hansen test of overid. restrictions (<i>p</i> -value)	0.660	0.737	0.600	0.683	0.667

Notes: ***, ** and * denote significance at the 1%, 5% and 10% levels, respectively. The standard errors in parentheses are corrected standard errors. Arellano–Bond AR(2) test: H_0 : No AR(2) in residuals. Rejection $\Rightarrow$ autocorrelation, model misspecification. Hansen test: H_0 : Instruments are valid. Rejection $\Rightarrow$ instruments may be invalid.

growth effects of education often shows that whilst tertiary attainment is positively associated with economic performance, the impact of secondary education is weak or even negative in advanced economies. Barro (1991) and Pritchett (2001) highlight that expansions in schooling, particularly at the secondary level, do not always translate into higher growth due to diminishing returns and skill mismatches. Similarly, Mankiw et al. (1992) and Lutz et al. (2008) find that in high-income regions, growth is more strongly driven by

tertiary human capital, with secondary schooling acting as a proxy for less-developed regions. Crespo Cuaresma (2010) and the OECD (2010) further emphasise that in Europe, secondary education is widespread but contributes little to productivity, as innovation and structural transformation rely mainly on tertiary-educated workers. This explains why regions with higher shares of secondary education often exhibit lower per capita GDP at the NUTS2 level. Per capita GFCF, as a proxy for investment in infrastructure, supports economic expansion by increasing capacity and efficiency. These findings align with growth literature, such as Barro (1991) and Mankiw et al. (1992), emphasising the importance of human capital, capital investment and structural factors in regional growth. Together, they underscore the need for policies to enhance employment, education, infrastructure and urbanisation for sustainable development.

The Major Components of CP Funds: The Role of ERDF and ESF

In Table 5, we use the same model structure as in Table 4 but focus on the ERDF¹¹ instead of total CP funding. Since ERDF is the greatest part of CP funds (Kah et al., 2020), its effects are expected to be like those of total CP funding, especially in terms of the direction of the results (Butkus et al., 2020).

Model 1 excludes the EQI¹² to address the concern about data coverage. Even without EQI, ERDF still shows a positive and significant effect, suggesting that the results are not only due to institutional quality. Models 2 to 5 include EQI and its interaction with ERDF. The positive and significant interaction terms show that ERDF works better in regions with higher institutional quality.

In Models 4 and 5, the dummy variables for PR regions have negative and significant coefficients, meaning these regions have lower baseline GDP levels compared to the rest or to PU regions. However, the interactions between ERDF and PR/IN are positive and significant, suggesting that ERDF funding is more effective in rural and intermediate areas, probably because these regions benefit more from investment.

In all models (1 to 5), the ER is strongly and positively linked with regional GDP per capita. This is in line with the ERDF goal of supporting job creation through innovation and investment projects. Other control variables, such as PD and private investment (measured by per capita GFCF), also have positive and significant effects except for education. These findings suggest that these factors play an important role in supporting regional economic development along with ERDF funding.

Now, our focus is on the ESF, the other relevant fund, which specifically aims at human capital development, better and incremental employment and promoting education and social inclusion (Bădîrcea et al., 2025).

Model 1 in Table 6 does not include the EQI¹³ and shows that ESF has a positive and statistically significant effect on regional GDP per capita. Even after accounting for institutional quality, ESF spending contributes positively to regional development. The results in Table 6 highlight the positive impact of the ESF and its interaction with EQI on regional economic performance in Models 2 to 4, emphasising the role of strong institutions in maximising the effectiveness of this type of fund (ESF) (Fritsch and Wyrwich, 2014).

¹¹As we described in the [Methodology](#) section and Appendix A.

¹²See footnote 9.

¹³See footnote 9.

Table 6: European Social Fund and Regional Economic Performance.

	<i>Model 1</i>	<i>Model 2</i>	<i>Model 3</i>	<i>Model 4</i>
$(t - 1)$ log of per capita GDP	0.760*** (0.0299)	0.712*** (0.0426)	0.441*** (0.0452)	0.350*** (0.0595)
ESF	0.00547** (0.0022)	0.0240*** (0.00152)	0.0446*** (0.00498)	0.0483*** (0.00781)
EQI		0.0456*** (0.00895)	0.0425** (0.0189)	0.0757*** (0.0255)
Interaction ESF*EQI		0.00325* (0.00195)	0.00613 (0.0393)	0.0312*** (0.00579)
LDR		−0.0801*** (0.0196)	−0.0478 (0.0396)	
Interaction ESF*LDR			0.129*** (0.0119)	
PR regions				−0.134** (0.0554)
Interaction ESF*PR				0.0143*** (0.00173)
IN regions				0.0510* (0.0279)
Interaction ESF*IN				0.00610 (0.0143)
Employment rate	0.175*** (0.0383)	0.381*** (0.0565)	0.376*** (0.0598)	0.415*** (0.0584)
Population density	0.0188*** (0.00369)	0.0346*** (0.00753)	0.0356*** (0.00779)	0.0315*** (0.00655)
Secondary education	−0.000819 (0.00920)	−0.0326* (0.0193)	−0.0346* (0.0192)	−0.0430** (0.0198)
Log of per capita GFCF	0.0996*** (0.0141)	0.239*** (0.0213)	0.241*** (0.0217)	0.232*** (0.0223)
Constant	2.142*** (0.299)	5.234*** (0.504)	5.334*** (0.494)	4.248*** (0.518)
Observations	3241	2091	2091	2091
Number of NUTS2 regions	238	179	179	179
Number of Instruments	35	56	55	56
Arellano–Bond test for AR(2) (<i>p</i> -value)	0.569	0.628	0.183	0.482
Hansen test of overid. restrictions (<i>p</i> -value)	0.852	0.268	0.200	0.206

Notes: ***, ** and * denote significance at the 1%, 5%, and 10% levels, respectively. The standard errors in parentheses are corrected standard errors. Arellano–Bond AR(2) test: H_0 : No AR(2) in residuals. Rejection $\Rightarrow$ autocorrelation, model misspecification. Hansen test: H_0 : Instruments are valid. Rejection $\Rightarrow$ instruments may be invalid.

Compared to PU regions, economic performance in PR regions is lower, reflecting inherent challenges. Other factors, such as employment and PD, consistently show positive effects, reinforcing the importance of the ESF focus on human capital and social inclusion.

Results suggest that institutional quality, labour market and human capital constraints are more significant barriers to regional economic performance than infrastructure gaps, particularly in LDR, PR and IN regions. Studies exist whose findings maintain that weak labour mobility, skill mismatches and low educational attainment hinder productivity growth more severely than physical infrastructure deficits in such regions (Iammarino

et al., 2019; McCann, 2020). According to Dijkstra et al. (2020), human capital shortfalls explain persistent underperformance despite substantial infrastructure investments. Similarly, Rodríguez-Pose and Wilkie (2019) find that institutional and labour market rigidities outweigh infrastructure in explaining lagging regional convergence.

III. Discussion

This study analyses the impact of the CP on regional economic performance across 240 NUTS2 regions of the EU27 over the period 2007–2022, with particular attention to the effects of heterogeneity across different regional typologies and institutional contexts. The descriptive analysis highlights persistent regional disparities in economic performance, which have become more pronounced in the years following the pandemic. This difference remains evident even when examining distinct regional types within the two groups of more developed and LDR: urban areas consistently record the highest GDP per capita, whilst predominantly rural and intermediate regions consistently demonstrate lower GDP per capita.

It is important to note that the average per capita payments from the CP are substantially higher in LDR compared to the overall sample, reflecting the greater allocation of Structural Funds to these regions to promote economic convergence. However, the significantly lower average scores of the institutional quality index (EQI) in LDR (−0.859) compared to the overall sample (−0.083) raise concerns. Indeed, this institutional quality deficit appears to undermine the effectiveness of CP implementation by limiting the capacity of local managing authorities to manage and carry out development projects. To maximise the impact of CP investments, financial transfers to lagging regions must increasingly be accompanied by governance reforms aimed at strengthening institutional quality.

Our results show that CP has a positive and significant impact on regional GDP per capita. This confirms that EU funds help boost regional growth, especially when combined with strong institutional quality. The interaction between CP and the institutional quality index suggests that CP works better in regions with better governance. This supports the idea that institutions matter for the success of public investment.

LDR have lower economic performance (negative and significant coefficient, Table 4, Model 3), aligning with their need for development support (Barca et al., 2012; González-Val, 2015). However, the positive impact of the interaction between CP payments and LDR suggests that the CP can help mitigate these disadvantages by addressing structural constraints and stimulating growth. LDR still face structural challenges that limit their growth, such as weaker institutions and lower investment levels.

By introducing additional layers of heterogeneity, we find that the dummy for predominantly rural regions (Table 4, Model 4) and its interaction with CP capture how territorial characteristics (urban vs. rural regions) influence the effectiveness of the policy.

In particular, predominantly rural regions exhibit lower average GDP but benefit more from the CP compared to urban areas. This suggests that targeted support in these regions is effective, especially when tailored to local needs.

Amongst CP funds, both the ERDF and the ESF positively influence regional growth. Their impact is stronger in areas with higher institutional quality (Tables 5 and 6),

underlining the need to improve governance for a better effectiveness of the different Structural Funds.

Control variables such as employment, PD and investments (GFCF) also support growth, whilst the effect of education is mixed, possibly due to skill mismatches in some regions. In conclusion, the CP helps reduce disparities, but its success depends on the quality of institutions, regional characteristics and targeted investments. Future policies should focus more on improving governance and tailoring support to local needs.

Conclusion

Although extensive research exists on the role of CP in addressing economic disparities, there remains a significant gap in understanding the detailed dynamics of CP, particularly the differential impacts of its components, such as the ERDF and the ESF, and the moderating effects of institutional quality along with economic performance across regional typologies.

Moreover, whilst existing literature often focuses on aggregated measures of regional economic performance, it tends to neglect the heterogeneity of impact across regions categorised by urbanisation levels (predominantly urban, rural and intermediate). Against this backdrop, this study examines the long-term relationship between CP, institutional quality and regional economic performance, using data from 240 NUTS2 regions across 27 EU member states from 2007 to 2022. Using a two-step system GMM approach, this article provides a comprehensive analysis of these relationships, offering new insights into the differential impacts of CP on regional economic performance.

The findings of this study reveal several critical dimensions of the relationship between CP, institutional quality and regional economic performance. The empirical results confirm that CP payments, whether examined in aggregate or disaggregated by their components, have a positive and significant impact on economic performance across the full sample of regions. However, the magnitude of this impact varies significantly across region types. In fact, economic performance is lower in LDR than in other regions. However, the positive impact of the interaction between CP payments and LDR suggests that the CP can help mitigate these disadvantages by addressing structural constraints and stimulating growth. Similarly, economic performance in predominantly rural regions is always lower than their counterparts. The results highlight the importance of CP in reducing the gap, addressing structural disadvantages faced by marginalised and rural regions, such as limited market access, lower productivity and weaker infrastructure.

Moreover, the analysis highlights the pivotal role of institutional quality in mediating the effectiveness of CP investments. Regions with higher institutional capacity and governance quality not only exhibit stronger overall economic performance but also derive greater returns from CP funds. This finding aligns with existing studies emphasising the necessity of robust institutional frameworks to enhance the absorptive capacity of public investments and ensure their efficient allocation and utilisation. Additionally, positive and significant interaction coefficients between CP payments and predominantly rural regions (and intermediate regions as well) suggest that the Structural Funds are particularly effective in addressing structural constraints and stimulating growth in these disadvantaged areas.

Several implications can be deduced from the findings of this study.

First, the heterogeneity of CP impacts across regions highlights the need for policy-makers to design and implement region-specific interventions that consider each region's unique structural and institutional characteristics. Less developed and rural regions require targeted policies that address their specific challenges, such as improving infrastructure, increasing market access and promoting innovation and productivity.

Second, strengthening institutional quality must remain a key priority for policy-makers at all levels of governance. Transparent and efficient governance frameworks are essential to maximise the effectiveness of CP investments and to ensure long-term, sustainable economic growth. To improve institutional and administrative quality, European Union regions – especially less developed ones – require significant support from both national and European governmental levels. A paradox exists: LDR (those with a GDP per capita below 75% of the EU-27 average) receive the largest share of Structural Funds, yet they are simultaneously expected to efficiently implement CP and use some of these funds to bolster their institutional capacity. Without adequate support for their efforts to improve institutional quality, these regions risk a 'vicious cycle'. In this cycle, poor institutional quality leads to inefficient resource use, keeps GDP per capita low, prevents the improvement of economic and social infrastructure, and ultimately hinders any advancement in institutional quality itself.

A key policy recommendation is that efforts to reduce regional disparities should focus on strengthening the absorptive capacity of LDR, including through capacity-building programs, technical assistance and reforms aimed at improving institutional and administrative efficiency. This requires that EU institutions complement the financial support provided to regions through the CP with targeted interventions aimed at enhancing institutional quality. Without such improvement, especially in less developed and predominantly rural regions, the achievement of CP objectives risks proving ineffective. Furthermore, greater dissemination and sharing of best practices from regions with higher institutional quality should be encouraged towards those facing greater challenges, whilst also considering the fundamental differences between urban and rural regional typologies.

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Data Availability Statement

The data supporting the findings of this study are openly available from Eurostat, for example, 'Regional Economic Accounts (reg_eco10)'. The datasets were accessed via <https://ec.europa.eu/eurostat>. All derived variables, code and methodologies used for analysis are fully described in the manuscript to ensure reproducibility.

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Appendix A: Variables and Data Sources

<i>Variable notation</i>	<i>Definition</i>	<i>Sources of the data</i>	
PCGDP	GDP per capita (in PPS) at NUTS2 level	Eurostat regional statistics	2007–2022
CP	The Cohesion Policy transfers, annual payment/Euro per capita.	European Commission (Eurostat), Cohesion Open Data Platform	2007–2022
EQI	The European Quality of Government Index (EQI). Government quality is defined as the impartiality of institutions, where public officials implement policies without bias. To measure this, government quality is divided into four components: rule of law, corruption, bureaucratic effectiveness, and democracy.	Quality of Government (QoG) Institute, University of Gothenburg.	2010, 2013, 2017, 2021
PCERDF	The European Regional Development Fund (ERDF), annual payment/Euro per capita. This fund assists all EU regions to reduce territorial discrepancies, investing heavily in several priority areas such as the digital economy, low carbon economy, innovation and research, SME competitiveness, and territorial co-operation amongst others.	European Commission (Eurostat), Cohesion Open Data Platform	2007–2022
PCESF	European Social Fund (ESF), annual payment/Euro per capita. This is primarily intended for the social component: investing in human capital, supporting citizens in finding jobs (inclusion of disadvantaged groups), offering opportunities on the labour market, sustaining education, creating businesses, and making public services more efficient amongst others.	European Commission (Eurostat), Cohesion Open Data Platform	2007–2022
LDR	Dummy_LDR = 1 represents less developed regions, EU member states that have a GDP < 75% of the EU27 average, mean value of 2007	Author's own calculation	2007–2022
Dummy_IN	Dummy_IN = 1 represents intermediate regions and 0 otherwise	Author's own calculation, please refer to Appendix B	2007–2022
Dummy_PR	Dummy_PR = 1 represents predominantly rural regions and 0 otherwise	Author's own calculation, please refer to Appendix B	2007–2022
Dummy_PU	Dummy_PU = 1 represents predominantly urban regions and 0 otherwise	Author's own calculation, please refer to Appendix B	2007–2022
ER	Employment rate, percentage (Y15–64) %	Eurostat regional statistics	2007–2022
PD	Population density (Inhabitants/km ²)	Eurostat regional statistics	2007–2022
PCGFCF	Per Capita Gross Fixed Capital Formation at Constant prices in EUR 2015	Eurostat regional statistics	2007–2022
SSA	Share of population with Secondary school attainment (25–64) (%)	Eurostat regional statistics	2007–2022

Appendix B: Classification of NUTS2 Regions Into PU, PR and IN

As a very first step, we downloaded the Local Administrative Units (hereafter LAUs) and NUTS2 shapefiles from the Eurostat website. These shapefiles provide spatial boundaries for LAU and NUTS2 regions across Europe. To determine how many LAUs lie within each NUTS2 region, a spatial join was conducted using the QGIS software. This spatial join enabled the association of LAU with their corresponding NUTS2 regions, establishing a basis for population analysis at the NUTS2 level.

In the next step, we cleaned the dataset to focus solely on NUTS2 regions. This involved extracting and keeping only those observations that corresponded to NUTS2 regions. The resulting data allowed for the analysis of the regional population and its distribution between rural and urban areas within each NUTS2 region.

We classified LAU as either rural or urban based on their population density. The threshold used is equal to 150 inhabitants per square kilometre, whereby LAU with a population density below this threshold were classified as rural, whilst those with a higher density were classified as urban. This classification reflects the degree of population concentration and forms the basis for further analysis of rural population distribution across NUTS2 regions.

To classify the NUTS2 regions into PU, IN and PR categories, the proportion of the population living in rural LAU relative to the total population of the NUTS2 region was calculated. For each NUTS2 region, the total population was added up, followed by the calculation of the share of the rural population (i.e., the population living in LAU classified as rural) relative to this total population. This rural population share provided the key metric for classifying each region.

Following OECD criteria for NUTS3 regions, based on the rural population share, each NUTS2 region was classified into one of three categories:

Predominantly Urban (PU): Regions where less than 15% of the population resides in rural LAU.

Intermediate (IN): Regions where between 15% and 50% of the population resides in rural LAU.

Predominantly Rural (PR): Regions where more than 50% of the population resides in rural LAU.

This classification framework adheres to established thresholds commonly used for distinguishing between urban, intermediate and rural areas at the regional level. The methodology ensures that the classification is grounded in quantitative measures of population distribution, enabling a robust analysis of regional characteristics (see Figure B1 for an overview of the classification process).

In comparing various studies that classify NUTS2 and NUTS3 regions into rural and urban categories, distinct methodological approaches emerge. Whilst our methodology classifies regions based on population density using a fixed threshold (150 inhabitants/km²) at the LAU level and aggregates them at the NUTS2 level, other studies introduce additional dimensions.

For instance, Gagliardi and Percoco (2017) use RDD to classify NUTS3 regions based on proximity to urban centres, focusing on how rural areas near cities benefit from agglomeration effects, whilst more dispersed rural regions face different dynamics. This methodology highlights geographical proximity as a key driver of regional development.

Similarly, Brezzi et al. (2011) use the OECD extended regional typology, categorising regions into Predominantly Urban (PU), Intermediate (IN) and Predominantly Rural (PR) but add a spatial accessibility criterion, distinguishing rural areas based on travel time to an urban centre with more than 50,000 inhabitants. This adds a geographic element to their classification that contrasts with our focus on population distribution.

Egri and Lengyel (2024) also follow Eurostat urban–rural classification but apply it at the NUTS3 level. Their classification is like ours but focuses more on the spatial distribution of population density at a different regional level. Tödtling and Trippel (2005) take a different approach by classifying regions based on their innovation systems and sectoral specialisation, distinguishing between low-tech rural regions and more urbanised areas. This classification is focused on economic structure and innovation potential, differing from our population-based methodology. Finally, Kashnitsky et al. (2021) also use Eurostat urban–rural framework, but their classification emphasises demographic trends, such as migration and ageing, alongside population density.

Figure B1: Methodology for Classifying NUTS2 Regions Based on Rural and Urban Population Distribution. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/jcms.70050)]

